

STEM Course Prep
MATLAB Exercises
Chapter 1

Slides 1-5

1. Open MATLAB. Change font/colors if needed.

Slides 6-8

2. Enter the following at the command prompt:

a = 2

Notice it repeats the value of 'a'. Look in the workspace – you will find the value for 'a', as well as the min/max of 'a'.

3. Enter the command:

a = 2;

Notice the value of 'a' is suppressed.

4. Enter the command:

a=a+7

followed on a new line by:

a=a*10

The final value should be 90 ((2+7)*10 = 90).

Slides 9-11

5. Press the up arrow key. All recent commands will be displayed one at a time.
6. Clear the variable 'a' with the following command:

clear a

Look in the workspace – 'a' is gone. The command 'clear' will clear all variables.

7. Enter the commands:

```
Radius = 2.37
```

```
Radius = radius+2
```

MATLAB is case-sensitive! Radius is defined; radius is not.

8. Enter the command:

```
who
```

MATLAB shows all defined variables.

9. Enter the command:

```
whos
```

MATLAB shows info about the variables – you can see if it's an integer/real number.

Change the variable 'Radius' into an integer. Enter the command:

```
Radius_integer = int32(Radius)
```

MATLAB rounds the value to 4.

10. Enter the command:

```
whos
```

Slides 12-15

11. Calculate the density of air at standard temperature using the ideal gas law

$$\rho = \frac{P}{RT}$$

if pressure (P) is 101300 Pa, gas constant (R) is 286.9, and temperature (T) is 293 K. Enter the command:

```
P =101300;  
R=286.9;  
T=293;  
Density= P/(R*T)
```

Result: 1.2051

Change to long format. Enter:

```
format long  
Density
```

Result: 1.205067230379590

To see other format types:

help format

12. Determine the final displacement after 30 seconds of a vehicle that has an initial velocity of 30 m/s and a deceleration of 1 m/s². Initial displacement is 27900 m. The kinematic equation is:

$$s = s_o + v_o t + \frac{1}{2} a t^2$$

```
initial_disp = 27900;  
initial_velocity = 30;  
time=30;  
accel = -1;  
final_disp = initial_disp+initial_velocity*time+(accel*time^2)/2
```

Result: 28350

Slides 16-21

13. Enter:

help elfun

Enter:

help sin

14. Calculate the cosine of 75°. We are calling the function cosd and 75 is the input argument.

cosd(75) used cosd because the input argument was in degrees

Result: 0.258819045102521

cos(75*pi/180)) to use COS the input argument must be in radians

Result: 0.258819045102521

Calculate the cosecant of 0.3 rad. Enter:

csc(0.3)

Result: 3.383863361824123

Solve $\cot^{-1}(2.51)$. Find answer in degrees and radians. Enter:

acot(2.51)

acotd(2.51)

Result: 0.379131807531353 (rad), 21.722652450712772 (deg)

15. Test out the various rounding functions: Use -10.456 as the test value.

a=-10.456

round(a)

ceil(a)

floor(a)

fix(a)

sign(a)

Result: -10 (standard rounding)

-10 (rounds towards positive infinity)

-11 (rounds towards negative infinity)

-10 (rounds towards zero)

-1 (-1 indicates value less than zero. 1 would be given for positive number)

16. The displacement for a mass in a second-order underdamped system can be found using the equation:

$$x(t) = e^{-\xi\omega_n t} \left\{ \left[\frac{\xi}{\sqrt{1-\xi^2}} x(0) + \frac{1}{\omega_d} \dot{x}(0) \right] \sin(\omega_d t) + x(0) \cos(\omega_d t) \right\}$$

where $\omega_d = \omega_n \sqrt{1 - \xi^2}$

Using the following values, calculate displacement.

$$\omega_n = 6 \text{ rad} \quad (\text{natural frequency})$$

$$\xi = 0.5 \quad (\text{damping ratio})$$

$$x(0) = 10 \text{ m} \quad (\text{initial displacement})$$

$$\dot{x}(0) = 0 \text{ m/s} \quad (\text{initial velocity} - \text{released from rest})$$

$$t = 0.605 \text{ s} \quad (\text{time})$$

```
clear
damp_ratio = 0.5;
w_n = 6;
init_disp=10;
init_vel = 0;
t=0.605;
w_d= w_n*sqrt(1-damp_ratio^2);
a=init_disp*cos(w_d*t);
b=exp(-damp_ratio*w_n*t);
c=(damp_ratio/sqrt(1-damp_ratio^2)*init_disp);
d=(1/w_d)*init_vel;
displacement=b*((c+d)*sin(w_d*t)+a)
```

17. Enter the commands:

3<5

Result: 1 (3 is less than 5 so the statement is true. True statements always result in 1.)

10>=10

9>=10

11>=10

Result:

1 (10 is equal to 10 – true statement – true is represented by a value of 1)

0 (statement is false. False statements result in a value of zero)

1 (statement is true)